

Web Development

Labo 10

1. Wat is Responsive Webdesign?

Responsive webdesign zorgt ervoor dat een website zich automatisch aanpast aan:

- **Kleine schermen** (smartphones)
- **Middelgrote schermen** (tablets)
- **Brede schermen** (laptops/desktops)

Een responsive site betekent:

Geen horizontale scroll

Tekst blijft leesbaar

Afbeeldingen schalen mee

Layout herschikt automatisch

Breakpoint	Class infix	Dimensions
X-Small	<i>None</i>	<576px
Small	<i>sm</i>	≥576px
Medium	<i>md</i>	≥768px
Large	<i>lg</i>	≥992px
Extra large	<i>xl</i>	≥1200px
Extra extra large	<i>xxl</i>	≥1400px

2. Viewport

De **viewport** = het zichtbare deel van je webpagina.

Belangrijkste meta-tag:

```
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Jari's Labo's</title>
<link rel="stylesheet" href="Labo_en_Verslagen/SiteFiles/CSS/styles.css">
```

Dit zorgt voor:

- width=device-width → breedte = breedte van het toestel
- initial-scale=1.0 → geen automatische zoom

Best practices:

- Geen horizontaal scrollen
- Geen vaste breedtes (bv. 1200px)
- Gebruik flexbox, grid of percentage-breedtes
- Gebruik **media queries** om lay-out te veranderen

3. Mobile First

Mobile First = eerst bouwen voor smartphones, daarna uitbreiden naar grotere schermen.

Opbouw:

1. **Basis CSS** = smartphone (géén media queries)
2. Extra CSS met media queries voor grotere schermen:
 - min-width: 768px voor tablet
 - min-width: 992px voor desktop

4 Responsive afbeeldingen

```
html {  
  min-height: 100%;  
  max-width: 100%;  
  overflow-x: hidden;  
  background: radial-gradient(ellipse at top, #2B1B28 30%, #000000 90%);  
  color: #FBE8C8;  
  background-size: cover;  
  background-repeat: no-repeat;  
}
```

5 Media Queries

Wat is een media query?

Een CSS-regel die alleen geldt als een bepaalde conditie waar is (vb: schermbreedte).

Basis syntax:

```
@media (min-width: 576px) {  
  .container-sm, .container {  
    max-width: 540px;  
  }  
}  
  
@media (min-width: 768px) {  
  .container-md, .container-sm, .container {  
    max-width: 720px;  
  }  
}  
  
@media (min-width: 992px) {  
  .container-lg, .container-md, .container-sm, .container {  
    max-width: 960px;  
  }  
}  
  
@media (min-width: 1200px) {  
  .container-xl, .container-lg, .container-md, .container-sm, .container {  
    max-width: 1140px;  
  }  
}
```

Breakpoint	Class infix	Dimensions
X-Small	<i>None</i>	<576px
Small	<i>sm</i>	≥576px
Medium	<i>md</i>	≥768px
Large	<i>lg</i>	≥992px
Extra large	<i>xl</i>	≥1200px
Extra extra large	<i>xxl</i>	≥1400px

6 Bootstrap 5

Bootstrap is een **CSS Framework** dat draait rond:

- GRID systeem
- Componenten (cards, buttons, navbars...)
- Mobile-first design
- Responsieve lay-out

7.1 Bootstrap gebruiken

```
<!-- Bootstrap CSS -->
<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css" rel="stylesheet">
</div>
</div>
<!-- Bootstrap JS -->
<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/js/bootstrap.bundle.min.js"></script>
</body>
</html>
```

Bootstrap Grid – Super belangrijk

Bootstrap gebruikt een **12-kolommen grid**.

Basisstructuur:

```
html

<div class="container">
  <div class="row">
    <div class="col">Kolom 1</div>
    <div class="col">Kolom 2</div>
  </div>
</div>
```

Responsive classes:

Class	Betekenis
col-12	Smartphone
col-md-6	Tablet
col-lg-4	Desktop

Voorbeeld: 1 (mobile) – 2 (tablet) – 4 (desktop)

```
<div class="col-12 col-md-6 col-lg-3">...</div>
```

```

<!DOCTYPE html>
<html lang="nl">
<head>
  <title>Bootstrap 5 </title>
  <meta name="viewport" content="width=device-width, initial-scale=1, shrink-to-fit=no">
  <!-- Bootstrap CSS -->
  <link rel="stylesheet" href="../CSS/bootstrap-grid.css">
  <!-- Eigen style -->
  <link href="../CSS/style.css" rel="stylesheet">
</head>
<body>
<div class="container-fluid">
  <div class="row">
    <div class="col-md-4">1</div>
    <div class="col-md-4">2</div>
    <div class="col-md-4">3</div>
  </div>
  <div class="row">
    <div class="col-md-2">4</div>
    <div class="col-md-8">5</div>
    <div class="col-md-2">6</div>
  </div>
  <article class="row">
    <div class="col-md-3"></div>
    <h1 class="col-md-7">Cursus CSS</h1>
    <div class="col-md-2"></div>
  </article>
</div>
</body>
</html>

```

```

* {
  box-sizing: border-box;
}

.container-fluid > div > div {
  background: linear-gradient(45deg, #00e1ff 0%, #002160 50%, #aa00ff 100%);
  padding: 20px;
  border-radius: 15px;
  color: white;
}

body {
  width: 100%;
}

html {
  min-height: 100%;
  max-width: 100%;
  overflow-x: hidden;
  background: radial-gradient(ellipse at top, #281828 30%, #000000 90%);
  color: #f8e8c8;
  background-size: cover;
  background-repeat: no-repeat;
}

```

Lorem ipsum dolor sit amet, consectetur adipisicing elit. Aspernatur, autem, cum deserunt distinctio doloremque dolores fugiat in iure labore magnam maiores nulla officiis quam ratione voluptas. Consequatur cupiditate, magni? Sed.

Lorem ipsum dolor sit amet, consectetur adipisicing elit. Debitis error et ipsa ipsum quaerat. Consequatur cum ea eveniet illo perferendis reiciendis tempora veritatis. Beatae delectus dignissimos eius, impedit repudiandae suscipit.

Lorem ipsum dolor sit amet, consectetur adipisicing elit. Animi blanditiis consectetur cumque distinctio, expedita, fugiat illum impedit maxime officia provident quo suscipit veritatis voluptate. Qui quis quisquam reiciendis sed voluptate!

Lorem ipsum dolor sit amet, consectetur adipisicing elit. A ab ad architecto, asperiores cupiditate delectus dignissimos eius eos eum expedita harum iste magni nobis quisquam quod rerum voluptatem. Distinctio, nam?

```

<!DOCTYPE html>
<html lang="nl">
<head>
  <title>Bootstrap 5 </title>
  <meta name="viewport" content="width=device-width, initial-scale=1, shrink-to-fit=no">

  <link rel="stylesheet" href="../../CSS/bootstrap-grid.css">
  <link href="../../CSS/style3.css" rel="stylesheet">
</head>

<body>

<div class="container">
  <div class="row g-0">
    <div class="col-3 " id="a"></div>

  </div>
  <div class="row g-0">
    <div class="col-3 " id="b"></div>
    <div class="col-9" id="c"></div>

  </div>
  <div class="row g-0">

    <div class="col-3 " id="d"></div>
    <div class="col-8 " id="e"></div>

    <div class="col-1 " >
      <div id="f"></div>
      <div id="g"></div>
    </div>

  </div>

</div>

```



```

1  #a {
2      border: solid black;
3      border-width: 0 8px 15px 0;
4      min-height: 165px;
5      background: rgb(255, 255, 255);
6  }
7
8  #b {
9      border: solid black;
10     border-width: 15px 8px 8px 0;
11     min-height: 190px;
12     background: rgb(255, 255, 255);
13 }
14
15 #c {
16     border: solid black;
17     border-width: 0 0 8px 8px;
18     min-height: 355px;
19     background: rgb(220, 40, 26);
20 }
21
22 #d {
23     border: solid black;
24     border-width: 8px 8px 0 0;
25     min-height: 150px;
26     background: rgb(0, 93, 159);
27 }
28
29 #e {
30     border: solid black;
31     border-width: 8px 0 9px 8px;
32     min-height: 150px;
33     background: rgb(255, 255, 255);
34 }
35
36 #f {
37     border: solid black;
38     border-width: 0 13px 8px 0;
39     min-height: 75px;

```

